FOR DEVELOPERS, DEVOPS & TECHNICAL ADMINISTRATORS

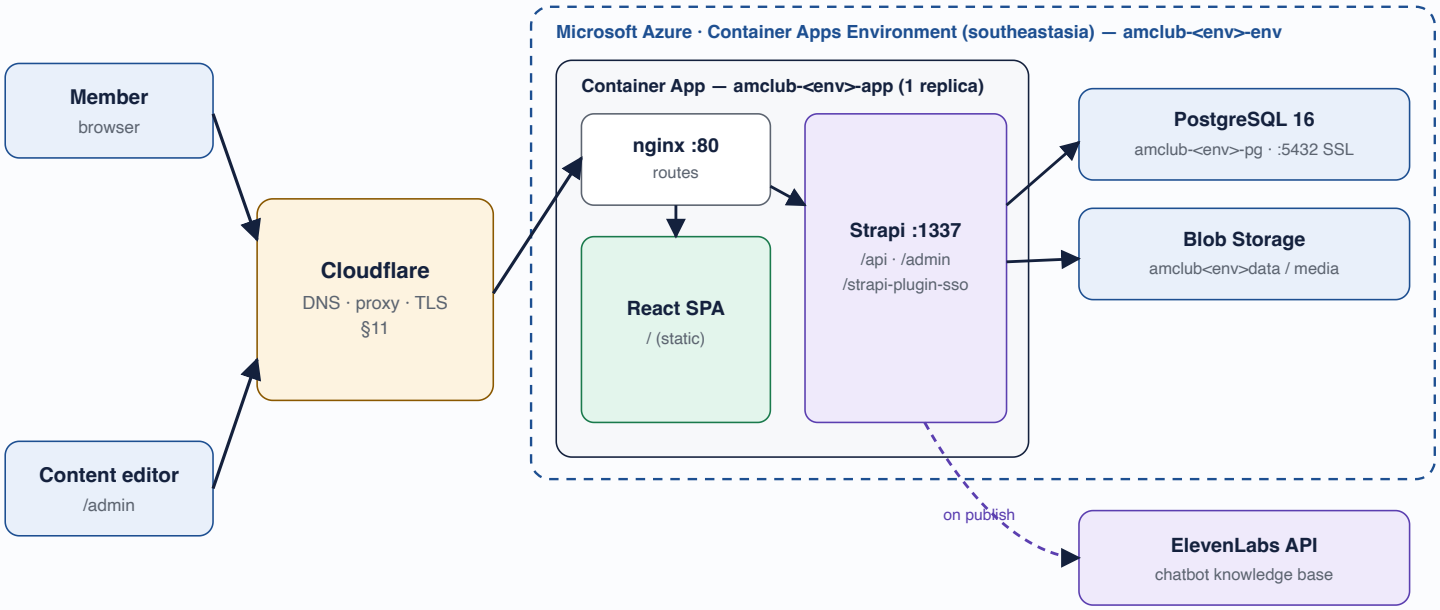
Technical Admin Guide
The American Club Website

Architecture, environments, deployment, CMS internals, content seeding, and operations runbook for the amclub-website monorepo. For day-to-day content editing, hand editors the CMS User Guide.

Last updated 10 July 2026 · Sources: repo `docs/`, `SPECS.md`, `infra/`, workflows · Screenshots from UAT admin

1. Architecture

Layer	Technology
Frontend	React 19 + Vite + Tailwind CSS v4 + TypeScript (strict), React Router v7
CMS	Strapi v5 (currently 5.46.x), PostgreSQL 16 in deployed envs, SQLite locally
Hosting	Azure Container Apps — one all-in-one container per environment (nginx + Strapi + built frontend)
Storage	Azure Blob Storage for media (`amclub<env>data` / container `media`)
IaC	Pulumi (TypeScript), state in Azure Blob backend (`azblob://state`)
CI/CD	GitHub Actions (`ci.yml` + `deploy.yml`)
Auth (admin)	Email/password + Microsoft Entra ID SSO via `strapi-plugin-sso`
Chatbot	ElevenLabs agent fed by a custom Strapi plugin syncing published content to its knowledge base



Runtime request flow. One all-in-one container per environment; the deployed SPA always calls its own origin (`/api`). `/uploads/*` is reverse-proxied by nginx to the blob container, so media serves from the site origin, not the raw blob host. DNS & TLS termination details are in §11.

Request routing (inside the container)

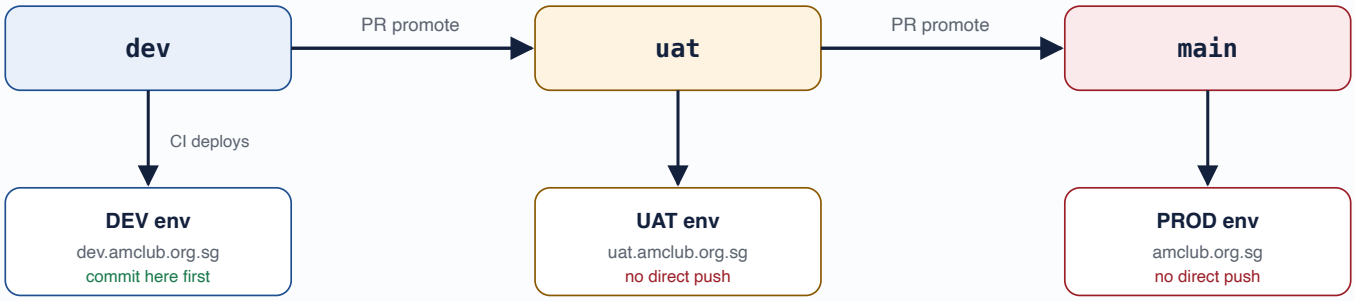
`infra/docker/nginx.conf` routes `/` → built SPA, `/api/*` → Strapi (:1337), `/admin/*` → Strapi admin, `/strapi-plugin-sso/*` → SSO endpoints, `/uploads/*` → Azure Blob container. The deployed frontend always talks to *its own origin* (`/api`) — never a hardcoded Strapi URL.

> **DESIGN & SPEC SOURCES OF TRUTH**
>
> `DESIGN.md` (root) holds the design tokens; `SPECS.md` (root) maps every page, component, and CMS binding. Read both before changing anything visual or structural. The Framer prototype is at `tactesting.framer.website`.

2. Environments & Branching

Three long-lived branches map 1:1 to environments. **All work lands on `dev` first**; promotion is via PR `dev → uat → main`. Never push directly to `uat` / `main`, never force-push them.

Branch	Environment	Public URL	Admin	Azure subscription / tenant
`dev`	DEV	dev.amclub.org.sg	dev.amclub.org.sg/admin	Prefix subscription (legacy tenant)
`uat`	UAT	uat.amclub.org.sg	uat.amclub.org.sg/admin	Client "Website" subscription (tenant `8137d2b4…`)
`main`	PROD	amclub.org.sg	amclub.org.sg/admin	



Three long-lived branches, three environments. Pushing to a branch triggers CI → deploy for that env (§9); promotion between environments is by pull request only. Never force-push `uat` / `main`.

Check what's queued for promotion:

```
git log --oneline origin/uat..origin/dev    # ready for UAT
git log --oneline origin/main..origin/uat   # ready for PROD
```

ENVIRONMENT-AWARE CODE ONLY

API base URLs, Strapi origins, feature flags, and analytics IDs come from env vars (`VITE_*` at build time for the frontend; Container App settings at runtime for Strapi). Never hardcode a single environment's URL in source.

CONTENT DIVERGENCE ON PROD

Content edited live on prod is logged under `content-updates/` (see `log.md` there) so it can be replayed onto dev seed scripts later. Don't edit dev content while doing prod content work.

3. Repository Layout

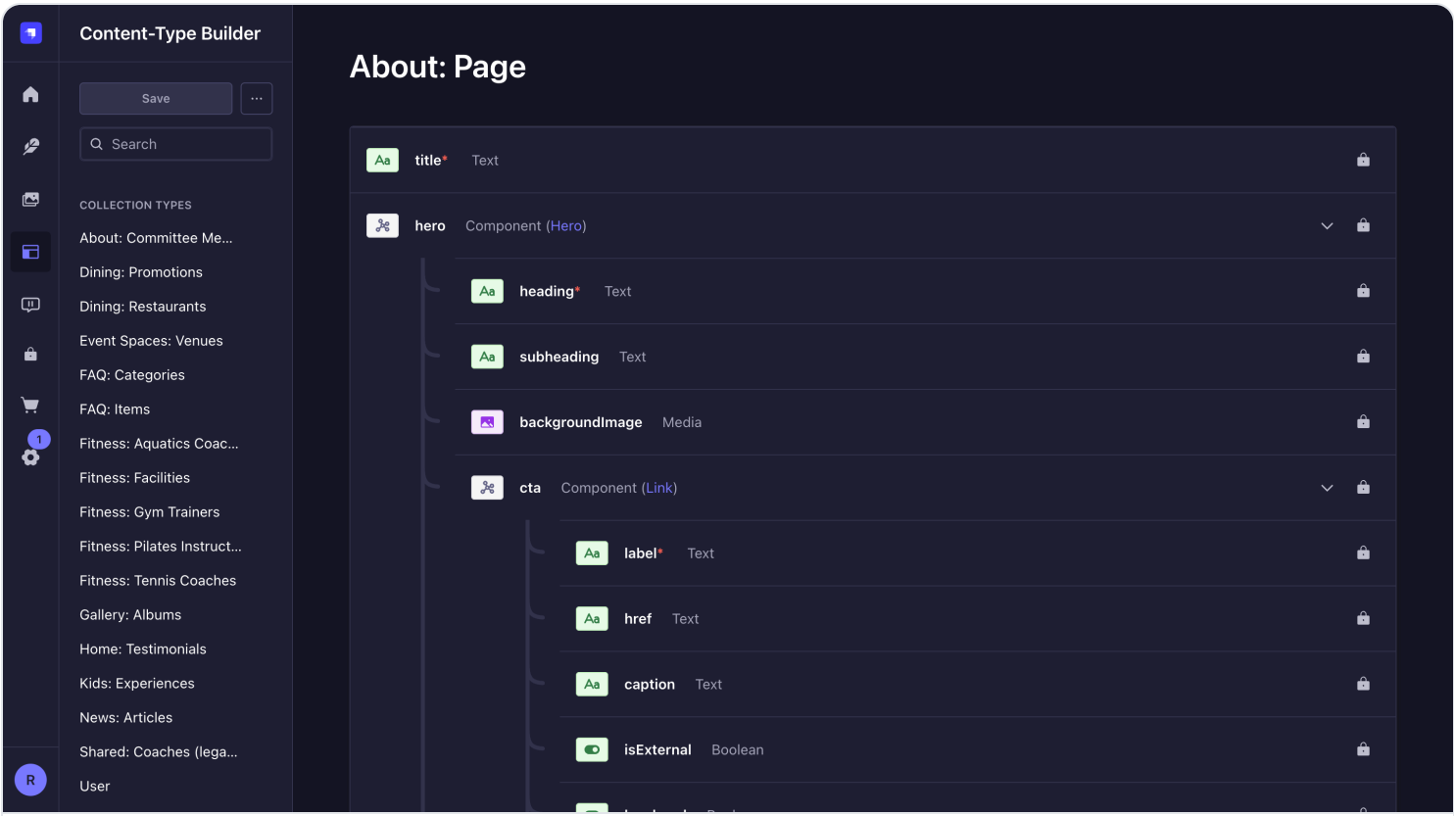
```
amclub-website/
├── cms/                  Strapi v5 project (APIs, plugins, providers, config)
├── frontend/             React + Vite + Tailwind SPA
├── infra/                Pulumi IaC (index.ts, Pulumi.<env>.yaml, README.md)
├── infra/docker/         Container entrypoint.sh + nginx.conf (Dockerfile is at repo root)
├── scripts/              seed-*.mjs / patch-*.mjs content scripts + seed-helpers.mjs
├── media/                ALL source assets, mirroring site structure (media/about/, media/dining/…)
├── content-updates/      Audit log of manual prod content edits (for replay)
├── docs/                 media-storage.md · strapi-patterns.md · troubleshooting.md · admin-guide/ (this)
├── .github/workflows/    ci.yml · deploy.yml
├── DESIGN.md             Design tokens (Stitch DESIGN.md spec)
└── SPECS.md              Page/component/CMS-binding source of truth
```

- **Conventions:** TypeScript strict, no `any`; named exports; 2-space indent, single quotes; conventional commits (`feat:`, `fix:`…).
- **Asset naming:** lowercase-with-hyphens only. Rename legacy spaced/uppercase files when touched and update every reference.

4. The Strapi CMS

Content types

~19 collection types + ~21 single types live under `cms/src/api/<type>/content-types/<type>/schema.json`. Admin display names follow the `<Section>: <Entity>` convention (e.g. *Dining: Restaurants*, *Global: Header*) — set in `info.displayName` only. **Never** change `singularName` / `pluralName`; they define API routes.



Content-Type Builder (read-only in deployed envs — schemas are locked because Strapi runs in production mode). Schema changes are made in code on `dev` and deployed.

SCHEMA CHANGE PROCEDURE

Edit `schema.json` (or use the builder against a *local* dev Strapi), commit to `dev`, let CI deploy. Also update `content-inventory.json` and `SPECS.md` in the same commit. Adding fields via the deployed admin is not possible (production mode) and would be lost anyway.

Plugins & customisations

Customisation	Location	What it does
`upload-azure-folders` provider	`cms/providers/upload-azure-folders/` (linked via `file:` dep)	Wraps the Azure upload provider; honours a `path` form-data field so blobs land in folder hierarchies. Must be COPYed in *both* Dockerfile stages.
`upload-path-to-folder` middleware	`cms/src/middlewares/upload-path-to-folder.ts`, registered after `strapi::body` in `cms/config/middlewares.ts`	Mirrors blob paths into Media Library folders on `POST /api/upload`.

Customisation	Location	What it does
Expiry filter	`cms/src/utils/expiry-filter.ts` + event / dining-promotion controllers	Hides expired events/promos from listing endpoints (fallback field e.g. `date`, explicit override `expiredAt`). Skipped when the query targets a slug/documentId so detail pages still resolve.
`elevenlabs-chatbot` local plugin	`cms/src/plugins/elevenlabs-chatbot/`	Syncs published entries to the ElevenLabs agent knowledge base; hourly cron (`expiryKbSweep` in `cms/src/index.ts`, rule `0 * * * *`) unsyncs expired rows. Needs `ELEVENLABS_API_KEY` / `ELEVENLABS_AGENT_ID`.
`clone-entry` local plugin	`cms/src/plugins/clone-entry/`	Adds the CLONE panel in the entry editor (new draft, `-copy` slug suffix; media/relations linked).
`strapi-plugin-sso`	npm dep, configured in `cms/config/plugins.ts`	"Continue with Microsoft" login via Entra ID (see §5).

KNOWN STRAPI V5 GOTCHA

Media fields *newly added* to existing single types/components don't persist via REST PUT on this project's local DB (e.g. `header.menuIcon`). Frontend fallbacks are used until proper migrations land — see `docs/strapi-patterns.md`.

ElevenLabs chatbot sync

The on-site assistant is an **ElevenLabs Conversational AI agent**. Its answers are grounded in a *knowledge base* (KB) that the local `elevenlabs-chatbot` plugin keeps in step with published CMS content. The plugin lives in `cms/src/plugins/elevenlabs-chatbot/`; host config is in `cms/config/plugins.ts`.

Mechanism	Detail
Publish-time sync	`autoSyncOnPublish: true` (env `ELEVENLABS_AUTOSYNC`). A lifecycle hook subscribes to *all* `api::*` models and, on publish/update of an **allow-listed** type, renders the entry to Markdown and upserts it as a KB document.
Allow-list	`DEFAULT_ELEVENLABS_CONTENT_TYPES` in `cms/config/plugins.ts` (all page singletons incl. the membership sub-pages + events, news, restaurants, **event spaces, fitness facilities, kids experiences, dining promotions**, committee members, FAQ items, testimonials, gallery albums). The plugin **Settings page** in `/admin` overrides this allow-list at runtime once

Mechanism	Detail
	saved. *Keep the list in step with content-type renames* — a stale UID (e.g. the old `api::venue.venue`) fails silently and that type simply never syncs.
Document naming	KB docs are prefixed `am-club:` (`ELEVENLABS_DOC_PREFIX`) so the sweep can identify and prune documents it owns without touching manually-added KB entries. The per-entry key is the **slug, falling back to `documentId`** — never the row `id`, which Strapi v5 regenerates on every publish (keying on it duplicated singleton docs each republish until fixed on 10 Jul 2026). Attached files become child docs named `<owner>:file:<name>`.
Markdown rendering	`markdown.ts` renders each entry with a titled source link (`Source: [Page Title](url)`) repeated at the end of *every* section — RAG retrieves chunks, so a single top-of-doc link never reaches the model. Rendered fields: summary scalars (name, capacity, location, phone, email…), `string` / `text` / `richtext` fields over 30 chars, components and dynamic zones. Citation URLs come from real SPA route maps in `sync.ts` (`COLLECTION_ROUTES` / `SINGLETON_ROUTES`).
RAG indexing	Creating a KB doc is *not enough* — each doc needs a RAG index (`POST /v1/convai/knowledge-base/{id}/rag-index`, then poll status via **GET**). Index *sequentially*; parallel POSTs spawn duplicate attempts. Each ElevenLabs account has a RAG storage quota (**20 MB** on the Club account; the ~2 MB figure was the old agency plan) — `rag_limit_exceeded` means quota, a plain `failed` at 100% usually means corrupt doc content.
Expiry sweep (cron)	`5 0 * * *` nightly at 00:05 **Asia/Singapore** (`expiryKbSweep` in `cms/src/index.ts`) unsyncs events / dining promotions whose date has passed — keeps the bot from citing stale events. "Today" is computed in SGT, not UTC: containers run UTC, so an ISO date would keep yesterday's events alive until 8am Singapore.
Excluded documents	Plugin Settings → **Excluded documents**. One pattern per line, matched case-insensitively against the upload's file name *and* its URL; `*` is a wildcard. A matching file is never pushed, and any doc a previous sync created for it is deleted on the next sync of its owner page — so the denylist is **retroactive**. **Why it exists:** ElevenLabs' PDF extractor flattens tables — a class timetable loses every row/column association and the agent then confabulates confident wrong pairings (verified on prod, 11 Aug 2026, gym class schedule). Text-only PDFs (menus, policies, forms) extract fine, so this is a targeted denylist, *not* a blanket "no PDFs" rule.
Teamup calendar sync	Plugin Settings → **Teamup calendar**. Pulls a rolling window (*days before* / *days after* today, SGT) from the Teamup API and pushes it to the KB. The calendar list loads automatically when the page opens; tick the subcalendars you want, or leave all unticked to pull everything. Repeating events are **collapsed to one document per series**: a 60-day window returns

Mechanism	Detail
	~1,900 occurrences but only ~230 series, so per-occurrence docs would flood the KB and bury one-off events. Cadence is derived from the *actual* gaps between occurrences (weekly / fortnightly / monthly / "every day"); when the spacing fits no clean pattern the real dates are listed instead of a cadence being invented. Sessions with differing times are listed individually rather than collapsed to one misleading time. **Past events expire automatically:** each run recomputes the window and deletes any Teamup doc that fell out of it — no separate cron needed. Use **Preview** first: it reports the window, event/series counts and sample rendered documents *without writing anything*.
Manual re-sync	The **"Sync to ElevenLabs"** panel on a published entry (`EditViewSidePanel`) forces an immediate upsert. Bulk re-sync is available from the plugin Settings page.
Widget config	The public widget reads agent/config from the plugin's `public-config` endpoint; `ELEVENLABS_AGENT_ID` is public, `ELEVENLABS_API_KEY` is a Container App secret.

REQUIRED ENV VARS

`ELEVENLABS_API_KEY` (secret) + `ELEVENLABS_AGENT_ID` (GitHub environment *variable*) per environment → injected by Pulumi. `TEAMUP_CALENDAR_KEY` + `TEAMUP_TOKEN` (both secrets, both optional) are required only when Teamup sync is enabled. Both live in env rather than on the Settings page: they are similar-looking opaque strings, and pasting the token into a "calendar key" field produced an empty calendar list with no obvious cause. Missing values do not fail the deploy — Teamup sync reports the missing variable by name when run. If unset, the plugin no-ops gracefully (no sync, no widget) — useful for local dev. Each environment points at its *own* ElevenLabs agent so UAT edits don't pollute the prod KB.

ACCOUNT TOPOLOGY (SINCE JULY 2026) — TWO ELEVENLABS ACCOUNTS

dev uses the agency's original ElevenLabs account (agent *American Club Agent - Dev*, `agent_2501kr…`) for internal testing. **uat** and **prod** use the *Club's own* ElevenLabs account (agents *amclub.org.sg - uat* `agent_6001kv…` and *amclub.org.sg - prod* `agent_8501kw…`) — so the two environments need the Club account's API key, and dev needs the old one. Keep this in mind when rotating `ELEVENLABS_API_KEY`: it is a *different key* for dev vs uat/prod. Prod's KB is the single source of truth; UAT may lag behind it.

AGENT PROMPT & TONE

The agent system prompt is versioned at `agent/prompt.md` — datetime-aware (the widget injects live Singapore date/time as dynamic variables each session), grounded-in-KB rules, department contacts,

> and tone aligned with `reference/TAC_Master_FAQ_Repository.xlsx`. Apply prompt changes to *all* agents via the ElevenLabs API/console after editing the file. Manually attached KB docs (no `am-club:` prefix, e.g. `manual:faq-master-repository`) are preserved by the sync — it only manages docs it owns.

Citations pipeline (page-title chips)

Three layers cooperate so every factual chatbot answer ends with numbered, page-titled source chips:

1. **KB docs** carry `Source: [Page Title](url)` lines in every section (`markdown.ts`; the manual FAQ doc `agent/faq-kb.md` carries one per Q&A).
2. **The prompt** makes the source link a required response-structure step and tells the agent to copy the titled markdown link verbatim (`More details: [Types & Joining Fees](…)`).
3. **The widget** (`ChatbotWidget.tsx`) lifts those lines into superscript footnote markers + chips labelled with the page title; internal links open *behind* the chat panel via SPA navigation, uploaded documents get a humanized filename label ("Capacity Chart (PDF)"), external links open a new tab.

> **AGENT LLM — KEEP IT ON CLAUDE**
> All three agents run `claude-sonnet-4-6`. This is deliberate: `gemini-2.5-flash` drops the per-turn citation rule after the first answer in multi-turn chats *regardless of prompt wording* (tested 1/5 → 2/5 after prompt tightening; Claude holds 5/5). If cost ever matters, trial `claude-haiku-4-5` before anything else — and re-run the eval (`scripts/eval-agent-faq.py`) after any model change.

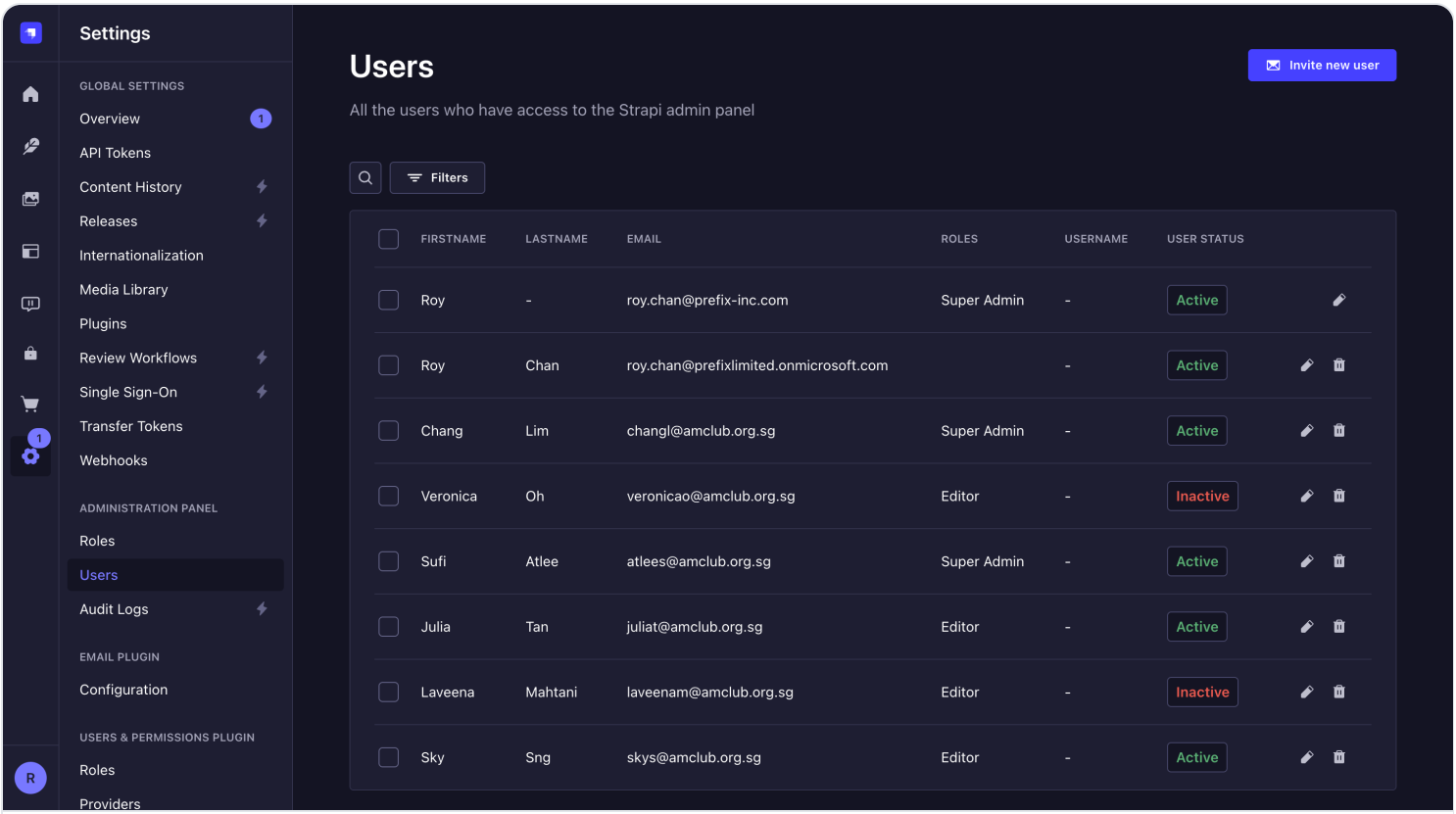
Full KB rebuild runbook

Needed after a plugin change that alters rendered markdown (all content hashes change), or when the KB is suspected stale. Order matters:

1. **Sweep** — *either* the plugin Settings page → "Sync All" (needs an admin session), *or* a no-op PUT of every allow-listed published entry via the content API with the seed token (fires the publish lifecycle; scripts follow the pattern in `scripts/` — sweep scripts are env-guarded so a dev script cannot hit UAT).
2. **Purge stale generations** — any doc named `am-club:*:id-<n>` after the sweep is a pre-fix leftover: detach from the agent, then delete. One dead doc id in the agent's KB list makes the whole attach PATCH 404, so the plugin validates per-doc with direct GETs (never the lagging search index).

3. **Re-index sequentially** — priority order (restaurants and page singletons first, gallery/news tail last). A changed doc = a *new* doc = its RAG index must be rebuilt; deleting the old doc frees its quota as you go.
4. **Verify live** — ask the deployed chatbot a KB-dependent question ("When is The 2nd Floor open today?") and confirm the answer carries a titled citation chip.

5. Admin Users, Roles & SSO



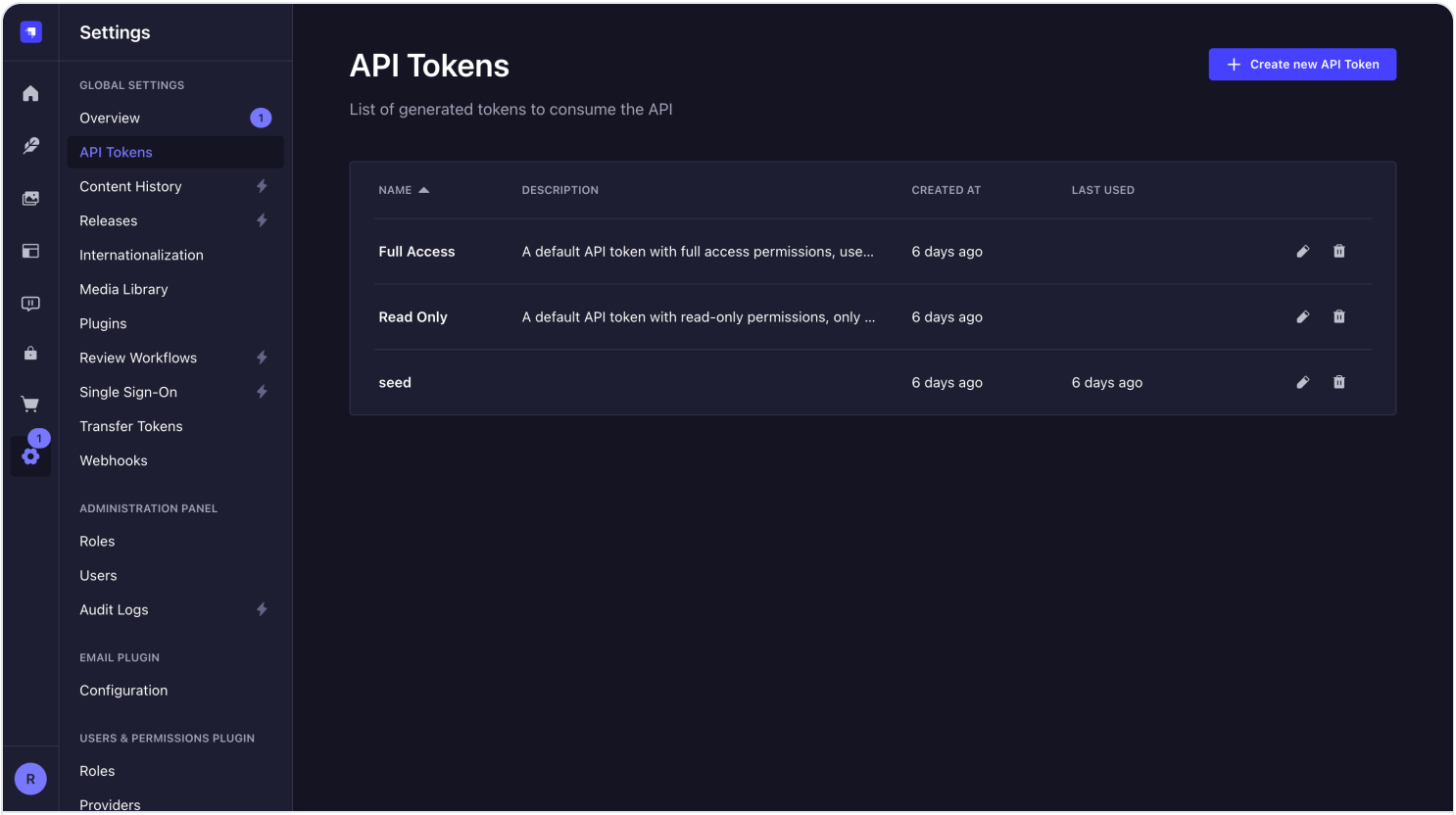
Settings → Administration Panel → Users. Club content staff get **Editor**; keep **Super Admin** to the maintenance team + one client owner.

- **Invite flow:** Settings → Users → *Invite new user* → assign role → send the registration link. Accounts show *Inactive* until the invite is accepted.
- **Roles:** Super Admin (everything), Editor (create/edit/publish all content), Author (own content only). Content staff = Editor.
- **SSO:** Entra app `amclub-strapi` (client id `a4f530c4…`). Redirect URI is derived from `PUBLIC_SITE_URL` per environment, so each env needs its redirect registered on the app. Configured via GH environment secrets `AZUREAD_TENANT_ID`, `AZUREAD_OAUTH_CLIENT_ID`,

`AZUREAD_OAUTH_CLIENT_SECRET`, `AZUREAD_SCOPE` — all optional; if unset, the env falls back to password login only.

> **SSO WHITELIST LOCKOUT**
>
> If `USE_WHITELIST: true` is enabled with an empty whitelist, SSO logins are rejected. Recover via password login → `/admin/settings/strapi-plugin-sso` → add emails, or revert the flag and redeploy. Details in `docs/troubleshooting.md`.

6. API Tokens



Settings → API Tokens. The `seed` token is what the content scripts authenticate with.

- Tokens are **per environment** and shown only once at creation — store them in the matching `cms/.env.seed.<env>` file (git-ignored). Never commit a token.
- The `seed` token needs full content + upload permissions for the seed scripts to upsert entries and upload media.

- Public reads from the frontend do *not* use a token — public role permissions on each content type's `find` / `findOne` govern that.

7. Seed Scripts & Content Ops

Pre-release content lives in seed scripts, not in ad-hoc `/admin` edits. Every piece of initial copy, image, or PDF must be reproducible against a fresh Strapi by re-running the scripts in `scripts/`.

Conventions

- `seed-<area>.mjs` — idempotent upserts for a page/collection (safe to re-run; matches by slug/key).
- `patch-<date>-<topic>.mjs` — dated one-off fixes, kept for audit/replay.
- `seed-helpers.mjs` — shared `api()`, `uploadFile()`, slugify, and the `TOP_LEVEL_BLOB_MAP` that maps `media/<section>/…` to blob paths.
- Auth/config from `cms/.env.seed.<env>`: `STRAPI_BASE_URL` + `STRAPI_API_TOKEN`.

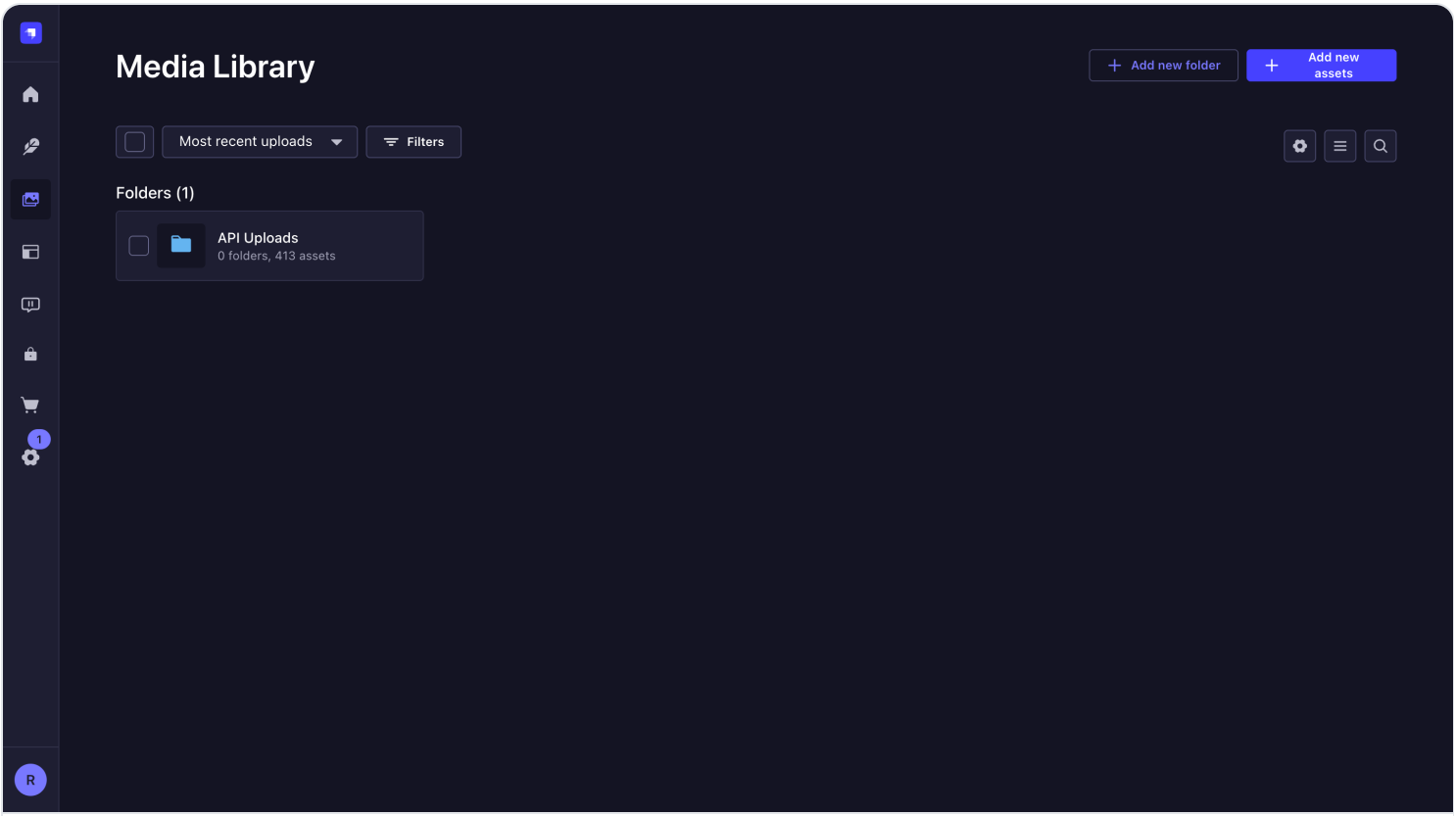
```
# run a seed against an environment
node scripts/seed-events.mjs --env=uat          # most scripts accept --env / --dry-run

# warm the container first if it scaled to zero (avoids mid-run fetch failures)
curl -sS "$BASE/api/upload/files?pagination[pageSize]=1" > /dev/null
```

Rules

- If you enter content ad-hoc in `/admin`, port it back into the matching seed script **in the same commit**.
- Manual *prod* content edits are logged in `content-updates/` (entry files + `log.md`) for later replay onto dev.
- Assets referenced by seeds live under `media/` mirroring site structure; delete orphans when replacing assets.

8. Media Storage



UAT Media Library. Seeded uploads currently land under "API Uploads" (the path→folder middleware mirrors blob paths when a `path` field is supplied).

- **Blob layout:** storage account `amclub<env>data`, container `media` (public read at Blob level), URLs like `https://amclubuatdata.blob.core.windows.net/media/uploads/<file>`.
- **Upload flow:** Strapi upload → custom `upload-azure-folders` provider → blob path from the `path` form-data field → `upload-path-to-folder` middleware mirrors the path into Media Library folders.
- **CORS:** admin thumbnails require a GET/HEAD/OPTIONS CORS rule on the blob service (set via Pulumi `BlobServiceProperties`; one-off fix via `az storage cors add`).
- Full details and incident history: `docs/media-storage.md`.

9. CI/CD

ci.yml — every push/PR to dev/uat/main

- CMS: `npm run build`
- Frontend: `npm run typecheck && npm run lint && npm run build`

- Infra: `npx tsc --noEmit`

deploy.yml — on push to a branch, after CI

1. Maps branch → GitHub environment (`dev` / `uat` / `prod`) and pulls that environment's secrets.
2. Builds the all-in-one Docker image (multi-stage: cms-builder → frontend-builder → runtime; `cms/providers/` copied in both CMS stages).
3. Runs `pulumi up` against the matching stack with `PUBLIC_SITE_URL` auto-set per branch, plus ElevenLabs + Entra SSO vars. The image is built *inside* `pulumi up` (`@pulumi/docker-build`) and pushed to the env's ACR as `amclub-app:latest`.

> **DOCKER BUILD CACHE IS DELIBERATELY OFF**
> The image builds with `noCache: true` (see `infra/index.ts`) — deploys are slower but guaranteed fresh, after a stale-layer bug once shipped a frozen admin bundle. `CMS_BUILD_NONCE` (= commit SHA) is kept in the Dockerfile so caching can be re-enabled safely later.

Secret / var	Purpose
`AZURE_CREDENTIALS`	Service principal JSON for the target subscription
`AZURE_STORAGE_ACCOUNT` / `AZURE_STORAGE_KEY`	Pulumi state backend (`azblob://state`). dev = `pulumistateb6e87494` (legacy tenant); uat + prod = `pulumistate56994511` (client tenant)
`ELEVENLABS_API_KEY` / `ELEVENLABS_AGENT_ID`	Chatbot knowledge-base sync
`AZUREAD_*`	Entra SSO (optional — password fallback if unset)

> **"DEPLOY SUCCEEDED" BUT THE SITE LOOKS STALE?**
> The new Container App revision may have failed its probes and traffic stayed on the old one. Check `az containerapp revision list` / `az containerapp logs show` — recipe in §14.

10. Infrastructure (Pulumi / Azure)

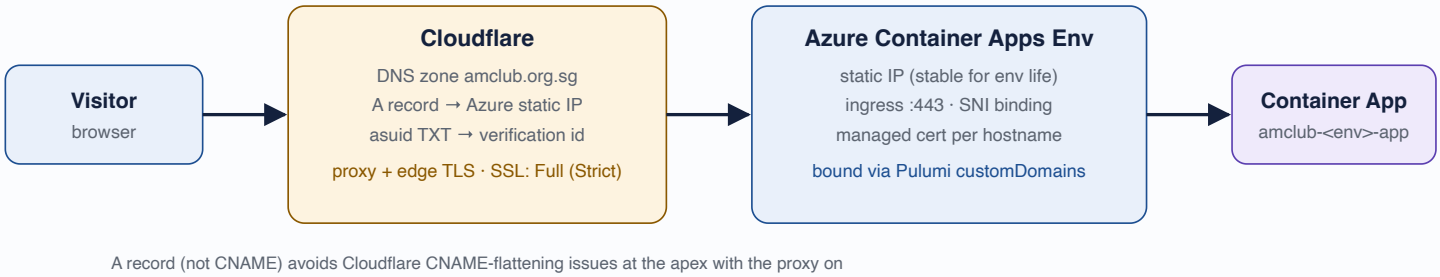
One Pulumi stack per environment (`infra/Pulumi.<env>.yaml`), all resources in Southeast Asia. Per stack:

Resource	Name pattern	Notes
Resource group	`amclub-<env>-rg`	
Container registry	`amclub<env>acr`	Basic SKU
PostgreSQL Flexible Server	`amclub-<env>-pg`	PG16, Standard_B1ms, 32 GB, 7-day backups
Storage account	`amclub<env>data`	Blob container `media` (public Blob read) + legacy file share
Container App	`amclub-<env>-app`	0.5 vCPU / 1 GB, scaling **fixed at min=max=1** — the container is stateful (Strapi + nginx in one image, uploads on an Azure Files volume at `/data`), so it must not scale horizontally. Keep ≥1 to avoid cold-start seed failures.
Container Apps env + Log Analytics	`amclub-<env>-env`	30-day log retention

- App secrets (DB password, Strapi keys/salts/JWT secrets) are auto-generated by Pulumi per deployment.
- Stack outputs: `appUrl`, `postgresHost`, `containerRegistryLoginServer`, `resourceGroupName`.
- Approx. running cost ≈ US$28/month per environment.
- **Direct DB access:** port 5432 is blocked from local machines — use Azure Cloud Shell or go through the content API instead.
- Setup walkthroughs and multi-subscription patterns: `infra/README.md`.

11. DNS & Custom Domains (Cloudflare → Azure)

DNS for `amclub.org.sg` is hosted at **Cloudflare**. Cloudflare resolves each hostname to the Azure **Container Apps Environment static IP** and (optionally) proxies the traffic. Azure terminates TLS at the Container App ingress with a **managed certificate per hostname**, bound via Pulumi. There is no separate web server or load balancer to manage.



DNS & TLS path. Cloudflare resolves the hostname to Azure's environment static IP; Azure presents the managed cert. Use an **A record** (not CNAME) — see `infra/index.ts` note on `containerEnvStaticIp`.

DNS records in Cloudflare

Per hostname there are two records — an address record and an Azure ownership-verification record:

Type	Name	Value	Proxy
`A`	`amclub.org.sg` / `www` / `uat` / `dev`	Container Apps Environment static IP — Pulumi output `containerEnvStaticIp`	Proxied (orange) in steady state; **DNS-only (grey) during cert issuance**
`TXT`	`asuid.<host>` (e.g. `asuid.uat`, `asuid` for apex)	Pulumi output `customDomainVerificationId` for that environment	n/a

```
# pull the two values Cloudflare needs, per environment
pulumi stack select <dev|uat|prod>
pulumi stack output containerEnvStaticIp        # → A record value
pulumi stack output customDomainVerificationId  # → asuid TXT value
```

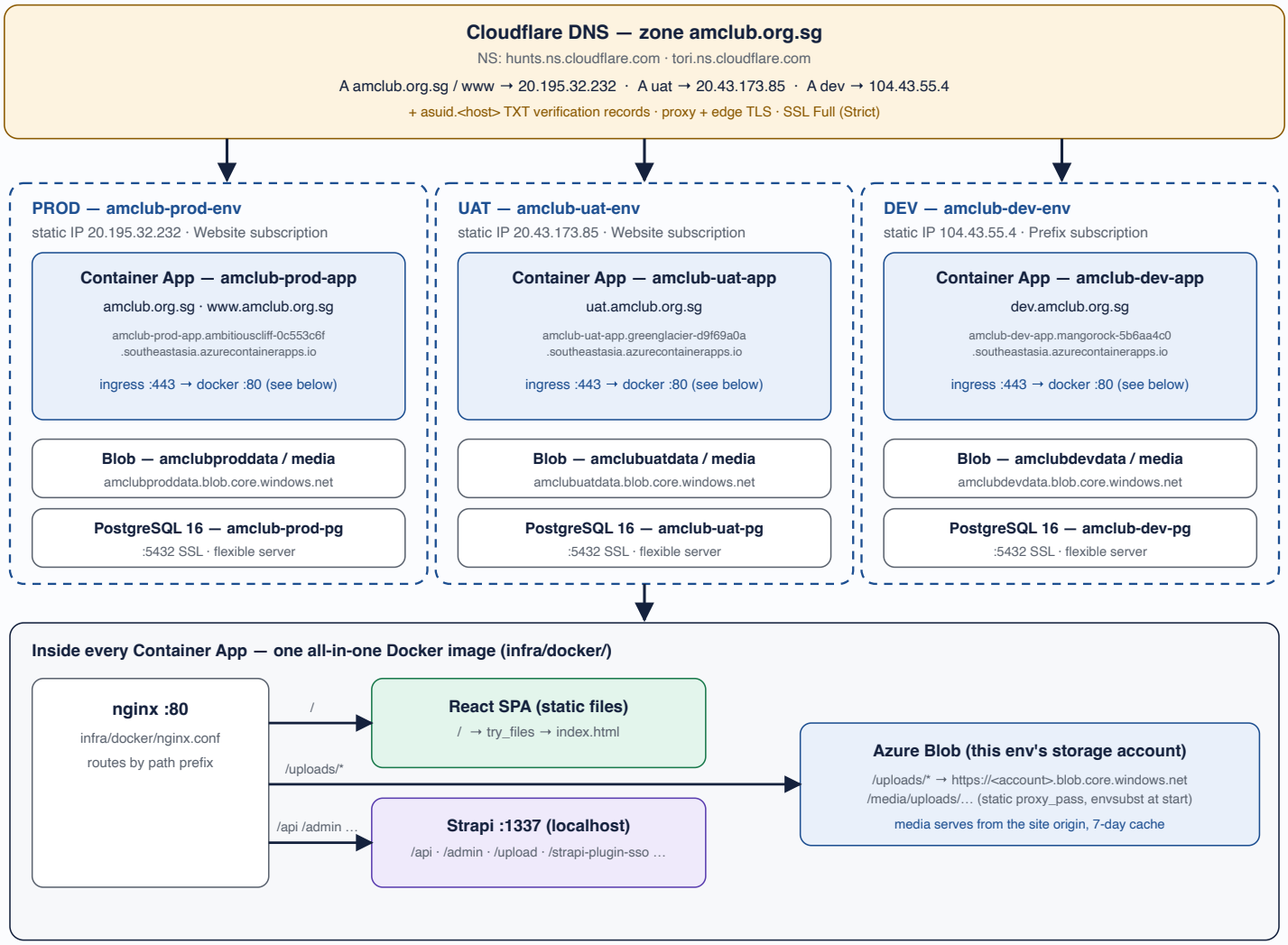
Live hostname → IP map

The zone's nameservers are `hunts.ns.cloudflare.com` / `tori.ns.cloudflare.com`. Current bindings (verified July 2026 — always re-check with `pulumi stack output` or `dig` before relying on these):

Env	Public hostname	A record → static IP	Container App ingress FQDN	Storage account
prod	`amclub.org.sg`, `www.amclub.org.sg`	`20.195.32.232`	`amclub-prod-app.ambitiouscliff-0c553c6f.southeastasia.azurecontainerapps.io`	`amclubproddata`
uat	`uat.amclub.org.sg`	`20.43.173.85`	`amclub-uat-app.greenglacier-d9f69a0a.southeastasia.azurecontainerapps.io`	`amclubuatdata`
dev	`dev.amclub.org.sg`	`104.43.55.4`	`amclub-dev-app.mangorock-5b6aa4c0.southeastasia.azurecontainerapps.io`	`amclubdevdata`

GOTCHA — THE CLUB'S INTERNAL NETWORK DOES NOT USE CLOUDFLARE

The American Club's internal office network resolves `amclub.org.sg` hostnames through the Club's **own internal DNS servers**, where the records were duplicated during the Cloudflare migration. Changes made in Cloudflare **do not propagate** to that internal DNS. If a static IP ever changes (an environment is recreated, moved to another subscription/region, or re-provisioned by Pulumi), the Club's IT team must **manually update the internal DNS records** — otherwise devices on the Club network keep resolving the old IP while the public internet works fine. Classic symptom: *"the site works on mobile data but not on Club Wi-Fi."* After any IP change: update Cloudflare, then notify Club IT with the new IP(s) from the table above.



Deployment topology & domain map (values verified July 2026). Every environment is the same shape: Cloudflare A record → Container Apps Environment static IP → one all-in-one container where nginx serves the SPA, proxies `/api` + `/admin` to Strapi on localhost, and reverse-proxies `/uploads/*` from that environment's blob storage account. dev lives in the Prefix subscription; uat and prod in the Club's Website subscription.

How the binding is wired (Pulumi)

Custom domains are declared in each stack's config and bound to the Container App ingress in `infra/index.ts` (SNI-enabled, referencing a managed certificate *by name*):

```
# infra/Pulumi.prod.yaml
amclub-website:customDomains:
  - name: amclub.org.sg
    certName: amclub.org.sg-amclub-p-260518090805
  - name: www.amclub.org.sg
    certName: www.amclub.org.sg-amclub-p-260519065509
```

WHY CUSTOMDOMAINS MUST LIVE IN STACK CONFIG

> Azure Container Apps ingress uses **PUT** (full-replace) semantics, not PATCH. If a hostname isn't in the desired-state body, *every* `pulumi up` *strips it* — the site loses its custom domain mid-deploy. Always add a new hostname to `Pulumi.<env>.yaml` **before** deploying; never bind domains only in the portal. (A prior `ignoreChanges` workaround failed — see `infra/index.ts` history.)

Managed TLS certificates

- One Azure-managed certificate **per hostname**, created in the Container Apps Environment. Naming pattern `<host>-amclub-<env-initial>-<timestamp>`.
- List / look up the `certName` to put in stack config:

```
az containerapp env certificate list -g amclub-<env>-rg -n amclub-<env>-env -o table
```

- Azure auto-renews managed certs. Renewal keeps the same cert resource, so `certName` in stack config does *not* change on renewal.

Runbook — add a new hostname (e.g. a vanity subdomain)

1. In Cloudflare, add the `A` record (→ `containerEnvStaticIp`) and the `asuid.<host>` `TXT` (→ `customDomainVerificationId`). Set the A record **DNS-only (grey cloud)** for now so Azure's validation reaches the origin.
2. Create the managed cert + add the hostname in Azure (portal: Container Apps Environment → Certificates, or via `az containerapp hostname add` then `az containerapp env certificate create`). Wait for validation to succeed.
3. Add `{ name, certName }` to `infra/Pulumi.<env>.yaml` → `customDomains`, commit to `dev`, promote as usual so the binding is captured in IaC.
4. Run the deploy (or `pulumi up`) and confirm the SNI binding.
5. Re-enable the Cloudflare **proxy (orange cloud)** and set **SSL/TLS mode → Full (Strict)** (Azure presents a valid public cert, so Strict is correct).

> **CLOUDFLARE PROXY VS. AZURE CERT VALIDATION**
> Azure's managed-cert issuance/renewal validates over HTTP/TLS to the hostname. With the Cloudflare proxy on, validation can fail because Cloudflare intercepts the request. If a new cert won't issue, temporarily switch the record to **DNS-only (grey)**, let Azure validate, then re-proxy. Renewals of *existing* certs are usually fine proxied, but keep this in mind if a renewal stalls.

CLOUDFLARE SETTINGS THAT MATTER

SSL/TLS: Full (Strict). **Always Use HTTPS:** on. **Apex:** use an A record to the static IP (not CNAME flattening). Keep Cloudflare caching conservative for `/admin` and `/api` (bypass) so CMS editing and API responses aren't cached; static SPA assets are safe to cache.

12. Frontend Integration

- **API client:** `frontend/src/lib/api.ts` — `fetchAPI<T>()` wrapper reading `VITE_API_URL` (defaults to same-origin `/api` in deployed builds). Strapi v5 populate syntax: `populate[field]=…` or `populate=*`.
- **Types:** `frontend/src/lib/types.ts` + `blocks.ts` mirror Strapi response shapes.
- **Pattern references:** `frontend/src/pages/VenueDetailPage.tsx` and `frontend/src/hooks/useHeaderData.ts` are the canonical CMS-wired examples.
- **Definition of Done** for any page: matching content type exists → component is props-only (zero hardcoded copy) → fetches via `fetchAPI` with loading/empty states → a published entry renders on the deployed URL. Track `cms-wired` status per page in `SPECS.md`.
- **Breakpoints:** mobile <768, tablet 768–1199, desktop 1200–1439, desktop XL 1440+.
- **SEO:** `frontend/src/lib/seo.ts` + `SeoManager` apply site-wide defaults (`site-config.siteName` + `defaultSeo`) on every route change; a page's own `shared.seo` component overrides *per field* (title, description, canonical, OpenGraph, Twitter). Pages call `usePageSeo(data?.seo)`. Defaults are seeded by `scripts/seed-site-config.mjs`.
- **Analytics:** `shared/Analytics.tsx` reads `site-config.googleTagId` (supports gtag `G-` / `GT-` / `AW-` and GTM `GTM-` IDs), injects the tag site-wide and emits SPA `page_view` events on route change. Empty field = analytics off.

HARD RULE

Never hardcode user-facing content (text, image URLs, CTA hrefs, list data) in `.tsx`. If real copy is going into a component, stop and add/extend a CMS field instead. Pages that "render fine offline" are regressions.

13. Verification Commands

```
cd frontend && npm run typecheck && npm run lint && npm run build
cd cms && npm run build
cd infra && npx tsc --noEmit
npm run test          # root: runs all of the above
```

Before any commit: run the relevant verification, then a browser regression sweep of the local build (every route at 1440px — no blank screens, no console errors), then targeted testing of the change itself.

14. Troubleshooting

Condensed from `docs/troubleshooting.md` — read that file for full symptom→cause→fix detail.

Symptom	Cause	Fix
Site works on mobile data but **not on Club Wi-Fi**	The Club's internal DNS still holds an old IP — it does not follow Cloudflare (see §11)	Ask Club IT to update the internal DNS records to the current static IPs (`pulumi stack output containerEnvStaticIp` per env)
Chatbot says it has **no information** about something that is on the website	Entry unpublished · field type not rendered · content type missing from the allow-list · KB doc exists but was never RAG-indexed	Check published + the data is in a real field/PDF; check the type is in `DEFAULT_ELEVENLABS_CONTENT_TYPES`; then re-sync and check the doc's rag-index status via GET (see §4 "Full KB rebuild runbook")
Chatbot answers but **without citation chips**	KB docs lack per-section `Source:` lines (pre-Jul-2026 render) — or the agent LLM was switched off Claude	Rebuild the KB so docs carry titled Source lines; confirm the agent LLM is `claude-sonnet-4-6` (gemini-flash drops the citation rule after turn 1)
`rag_limit_exceeded` when indexing KB docs	Account RAG storage quota (~2 MB) exhausted — usually by stale duplicate docs (`am-club:*:id-N` generations) or orphans still holding indexes	Purge stale `id-N` docs (detach → delete), drop indexes of unattached orphans, then re-index priority-first. A plain `failed` status instead means corrupt doc content — inspect the doc text.

Symptom	Cause	Fix
Broken thumbnails in admin Media Library	Missing CORS rule on the blob service	`az storage cors add --account-name amclub<env>data --services b --methods GET HEAD OPTIONS --origins '*' --allowed-headers '*' --exposed-headers '*' --max-age 3600`
Uploads appear flat under "API Uploads"	`upload-path-to-folder` middleware not registered (or no `path` sent)	Confirm `{ name: 'global::upload-path-to-folder' }` sits after `strapi::body` in `cms/config/middlewares.ts`
`Cannot find module 'upload-azure-folders'` at boot	`cms/providers/` missing from a Dockerfile stage	COPY providers before `npm ci` (builder) and after node_modules (runtime)
Deploy "succeeded" but site serves old build	New Container App revision unhealthy; traffic pinned to previous revision	`az containerapp revision list -n amclub-<env>-app -g amclub-<env>-rg` then `az containerapp logs show … --tail 200`
Seed script dies mid-run with `fetch failed`	Cold start after scale-to-zero	Keep min replicas = 1; warm with a trivial GET before seeding
Locked out of admin after enabling SSO whitelist	`USE_WHITELIST: true` with empty whitelist table	Password login → SSO settings page → add emails; or revert flag + redeploy
Pulumi state errors / NXDOMAIN	Wrong state storage account for the env	dev → `pulumistateb6e87494`; uat/prod → `pulumistate56994511`. Verify with `nslookup <account>.blob.core.windows.net` and `infra/Pulumi.<env>.yaml`
Custom domain dropped / site on the `*.azurecontainerapps.io` URL after a deploy	Hostname missing from `Pulumi.<env>.yaml` `customDomains`; PUT semantics stripped it	Add the `{ name, certName }` back to stack config and redeploy. See §11

Symptom	Cause	Fix
New managed cert won't validate / stuck "Pending"	Cloudflare proxy intercepting Azure's HTTP validation, or `asuid` TXT missing/wrong	Set the A record to **DNS-only (grey)**, confirm `asuid.<host>` TXT = `customDomainVerificationId`, retry, then re-proxy
Browser shows a TLS/SSL error on the custom domain	Cloudflare SSL mode mismatch or cert not yet bound	Set Cloudflare SSL/TLS to **Full (Strict)**; verify the SNI binding exists on the Container App ingress
Editor reports change not visible	Usually workflow, not infra	Walk them through the CMS User Guide §14 first (publish vs save, environment mix-up, expiry filter)

> **KEEP THE RUNBOOK ALIVE**
> Every new gotcha or pattern goes into `docs/troubleshooting.md` / `docs/strapi-patterns.md` when you land the fix — the next engineer (or session) saves hours.

15. Key File Reference

Purpose	Path
Frontend API client	`frontend/src/lib/api.ts`
Frontend types for Strapi shapes	`frontend/src/lib/types.ts`, `frontend/src/lib/blocks.ts`
Router / pages	`frontend/src/App.tsx`, `frontend/src/pages/`
Header data hook (CMS-wired example)	`frontend/src/hooks/useHeaderData.ts`
CMS plugin/provider config	`cms/config/plugins.ts`
CMS middleware order	`cms/config/middlewares.ts`
Custom upload provider	`cms/providers/upload-azure-folders/`

Purpose	Path
Path→folder middleware	`cms/src/middlewares/upload-path-to-folder.ts`
Expiry filter	`cms/src/utils/expiry-filter.ts`
Cron (KB sweep)	`cms/src/index.ts`
Local plugins	`cms/src/plugins/elevenlabs-chatbot/`, `cms/src/plugins/clone-entry/`
Infra entry point	`infra/index.ts` (+ `infra/README.md`)
Custom-domain / cert binding	`infra/Pulumi.<env>.yaml` (`customDomains`) · `infra/index.ts` (binding + `containerEnvStaticIp` / `customDomainVerificationId` outputs)
Seed helpers	`scripts/seed-helpers.mjs`
Docker	`Dockerfile` (root), `infra/docker/entrypoint.sh`, `infra/docker/nginx.conf`
Deep-dive docs	`docs/media-storage.md` · `docs/strapi-patterns.md` · `docs/troubleshooting.md`
Design tokens / page specs	`DESIGN.md` · `SPECS.md`